

DS 2500 *Fall* 2025 Lab Exercise 04

Deadline : 09/29/2025 11:59 PM EST

Late Deadline: 10/06/2025 11:59 PM EST with 50 points off

Total Points: 200 pts

Grading:

- This lab assignment is auto-graded. Gradescope can be a little picky, so make sure you don't put extraneous characters or whitespace in your answers -- and double-check the correct/incorrect confirmation.

Submission Instructions:

- Submit the Python file (`.py`) containing your solutions to Gradescope by the deadline specified at the top.
- You're allowed *unlimited* submissions.
- There are two deadlines. Submissions will not be accepted after the late deadline.

Disclosure: We've adapted lab problems from DS 2500 Spring 2025, so one or more problems may be the same or similar to those from that course.

Note: This lab assignment is intended to help you prepare for Quiz 1. Don't forget to review the additional problems in the lecture slides.

Problem 1 (50pts)

- Write a function `problem1()` that takes a list and returns the last element in the list. If the list is empty, return the string "empty list".

Example:

```
problem1([1,2,3])
# ^ returns 3
problem1(["1","2","3"])
# ^ returns "3"
problem1([])
# ^ returns "empty list"
```

Problem 2 (50pts)

- Write a function `problem2()` that takes a number and uses a `while` loop to generate a string containing the numbers from 1 up to that number, in order.

Example:

```
problem2(3)
# ^ returns "123"
problem2(5)
# ^ returns "12345"
```

Problem 3 (50pts)

- Write a function `problem3()` that takes two lists of strings and uses a `nested for loop` to generate a list where each element from the first list is combined with all elements of the second list joined together.

Example:

```
problem3(["0","1"], ["0","1"])
# ^ returns ["001", "101"]
problem3(["t","p"], ["e","a"])
# ^ returns ["tea", "pea"]
```

Problem 4 (50pts)

- Write a function `problem4()` that takes a list of tuples and returns a new list sorted by the second element of each tuple in ascending order.

Example:

```
problem4([("a", 100), ("b", 50), ("c", 200)])
# returns [("b", 50), ("a", 100), ("c", 200)]
problem4([("a", 1), ("b", 2), ("c", 3)])
# returns [("a", 1), ("b", 2), ("c", 3)]
```

Extra Problem 1 (0 pt; not graded)

- Write a function `extra_p1()` that takes a list of numbers and uses a `for loop` to generate a string containing the numbers in order.

Example:

```
extra_p1([1,2,3])
# ^ returns "123"
extra_p1([2,4,3])
# ^ returns "243"
```

Extra Problem 2 (0 pt; not graded)

- Define a class `Counter()` with an initializer that sets an initial value (i.e., store the attribute `value`).
- Implement an `increment(k)` method that adds `k` to the current value.
- Create an instance with initial value 1, call `increment` twice with values 10 and 100, and print the final value.

Example:

```
count = Counter(1)
count.increment(10)
count.increment(100)
print(count.value)
# ^ Upon proper implementation, you should get 111
```

Extra Problem 3 (0 pt; not graded)

- What will be printed after running the code below?

```
numbers = [1, 2, 3]
data = [10, 20, 30]
def modify_lists():
    numbers.append(4)
    data = [40, 50, 60]
    data.append(70)
modify_lists()
print(numbers)
print(data)
```

Extra Problem 4 (0 pt; not graded)

1. True or False: A function can have default values for parameters.
2. True or False: A function cannot be called inside another function.
3. True or False: A function must return one or more values.
4. True or False: Lists are mutable.
5. True or False: Lists can only hold one data type at a time.
6. True or False: Two or more variables can be assigned to the same list.
7. True or False: List cannot be accessed through negative indices. (indices: plural for index)
8. True or False: Dictionaries are immutable.
9. True or False: A for loop and a while loop are essentially the same if we use a conditional statement and a break keyword inside the for loop.
10. True or False: List comprehension is another technique that can replace a for loop.
11. True or False: Variables can change their value over time according to the instruction in a program.
12. True or False: Dictionary can include the same value several times, but cannot include the same key several times.
13. True or False: Tuples are mutable.
14. True or False: In a nested loop, the inner loop goes through all of its iterations for every single iteration of the outer loop.

